5	(a)	Faraday's law of electromagnetic induction states the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage.		1
	(b)	(i)	Induced e.m.f. = $(\Delta B)AN/(\Delta t)$ $= (2.0 \times 0.19 \times 1.5 \times 10^{-4} \times 120)/(0.13)$ [C1] = 0.053 V [A1]	1 1
		(ii)	Reading on voltmeter connected to coil C/ V: 0 0.053 0 [A1] (all three values required) Reading on voltmeter connected to Hall probe/ V: Zero in middle column [B1] Final column correct sign {positive} [B1] Final column correct magnitude (0.20) [B1]	1 1 1 1